

vector

```
int main()
{
    vector< int > v1( 3 );
    vector< int > v2( 6 );

    cout << "Size of v1 is " << v1.size() << "\nv1:";
    output( v1 );

    cout << "Size of v2 is " << v2.size() << "\nv2:";
    output( v2 );

    for( size_t i = 0; i < 3; i++ )
        v1[ i ] = i + 1;

    for( size_t i = 0; i < 6; i++ )
        v2[ i ] = i + 4;

    cout << "v1:";
    output( v1 );
    cout << "v2:";
    output( v2 );

    vector< int > v( v1 );

    cout << "Size of v is " << v.size() << "\nv:";
    output( v );
}
```

```

    if( v1 != v2 )
        cout << "v1 != v2\n\n";

    cout << "v1 = v2:\n\n";
    v1 = v2;

    cout << "v1:";
    output( v1 );
    cout << "v2:";
    output( v2 );

    if( v1 == v2 )
        cout << "v1 == v2\n\n";

    cout << "v1[ 3 ] is " << v1[ 3 ] << endl << endl;

    v1[ 3 ] = 100;

    cout << "v1[ 3 ] is " << v1[ 3 ] << endl << endl;

    cout << "v1:";
    output( v1 );
}

```

Size of v1 is 3

v1: 0 0 0

Size of v2 is 6

v2: 0 0 0 0 0 0

v1: 1 2 3

v2: 4 5 6 7 8 9

Size of v is 3

v: 1 2 3

v1 != v2

v1 = v2:

v1: 4 5 6 7 8 9

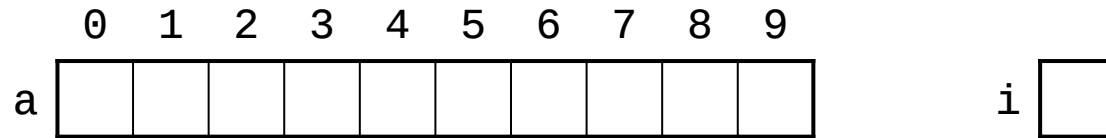
v2: 4 5 6 7 8 9

v1 == v2

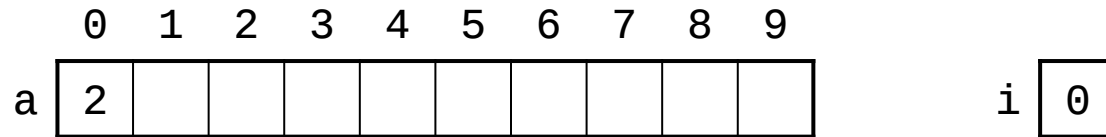
```
v1[ 3 ] is 7
```

```
v1[ 3 ] is 100
```

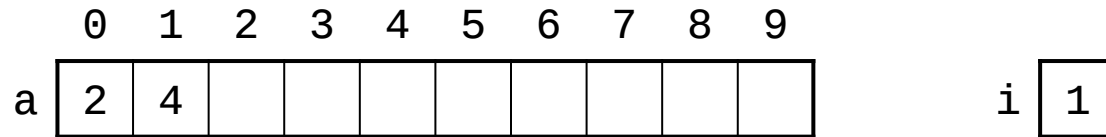
```
v1:    4    5    6 100    8    9
```



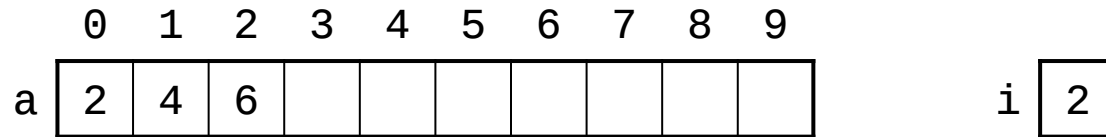
```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}
```



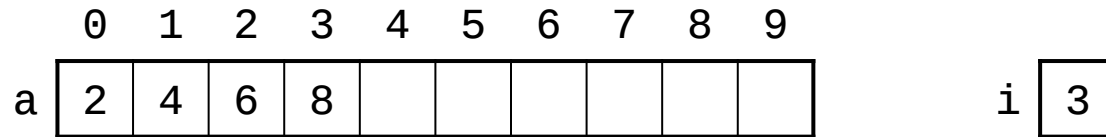
```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}
```



```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}
```

```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}
```



```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}
```

	0	1	2	3	4	5	6	7	8	9	
a	2	4	6	8	10						i 4

```

int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}

```

```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ];
    for( int i = 0; i < arraySize; i++ )
        a[ i ] = 2 + 2 * i;
}
```

```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```

```
class vector
{
    vector( unsigned int n = 0 )
    {
        mySize = n;
        myFirst = new int[ n ]();
    }

    vector( const vector &x )
    {
        mySize = x.mySize;
        myFirst = new int[ mySize ];
        for( unsigned int i = 0; i < mySize; i++ )
            myFirst[ i ] = x.myFirst[ i ];
    }

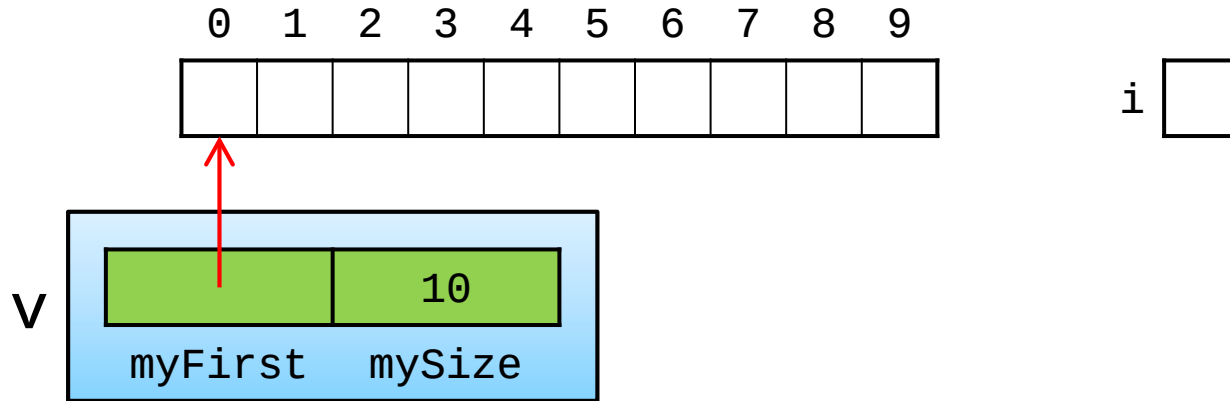
    unsigned int size()
    {
        return mySize;
    }
}
```

```
void push_back( const int val )
{
    resize( mySize + 1 );
    myFirst[ mySize - 1 ] = val;
}

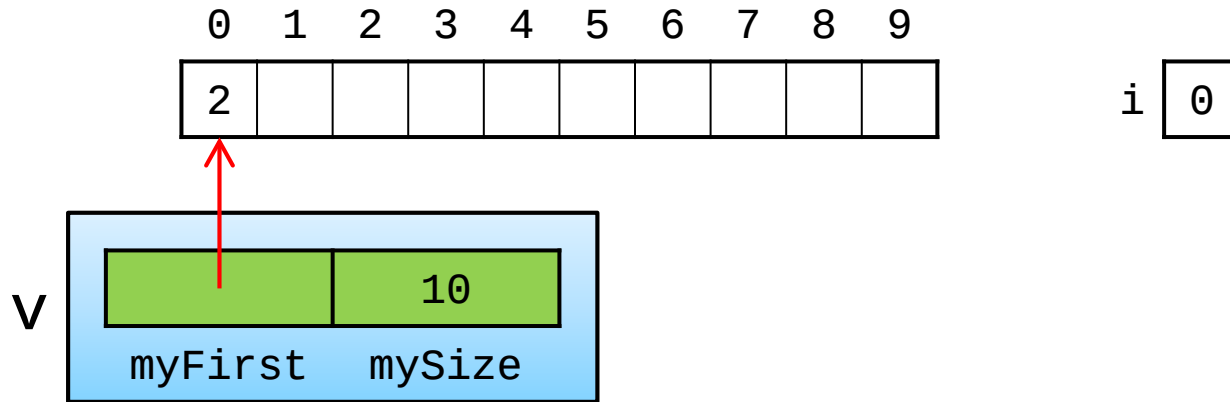
void pop_back()
{
    if( mySize > 0 )
    {
        --mySize;
        myFirst[ mySize ] = 0;
    }
}

void resize( unsigned int n )

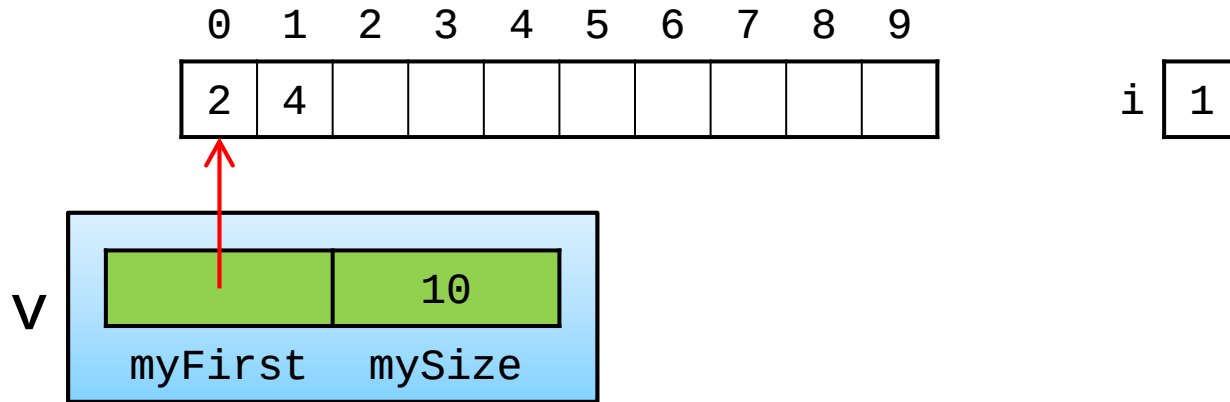
unsigned int mySize = 0;
int *myFirst = nullptr;
};
```



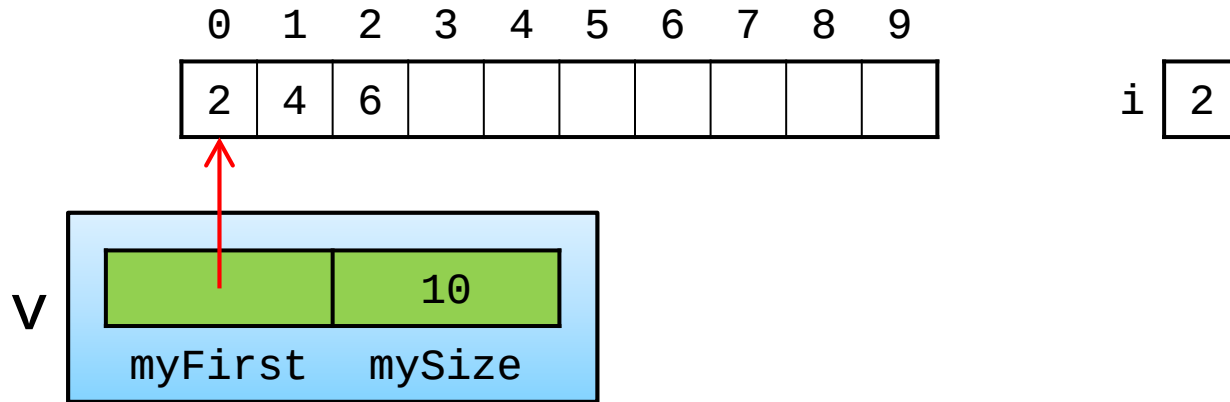
```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    cout < v.size() << endl;
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```



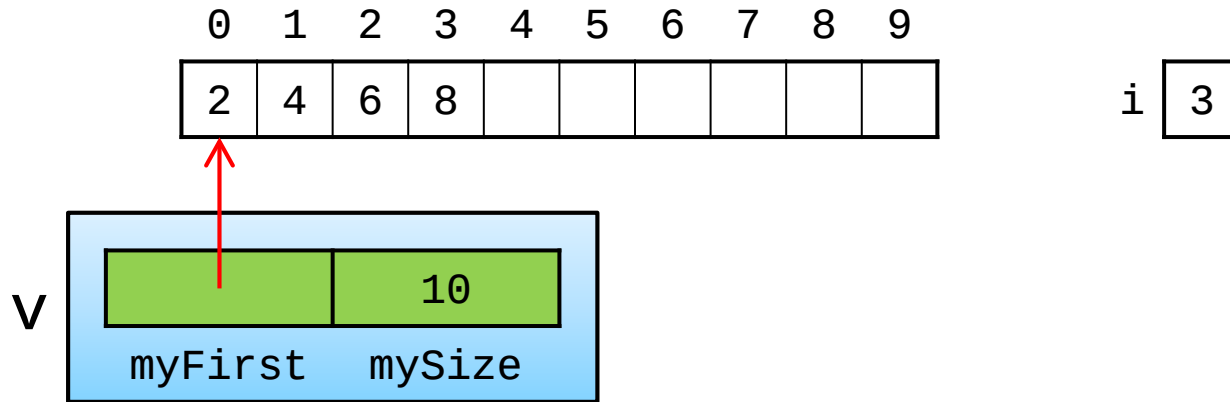
```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    cout < v.size() << endl;
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```

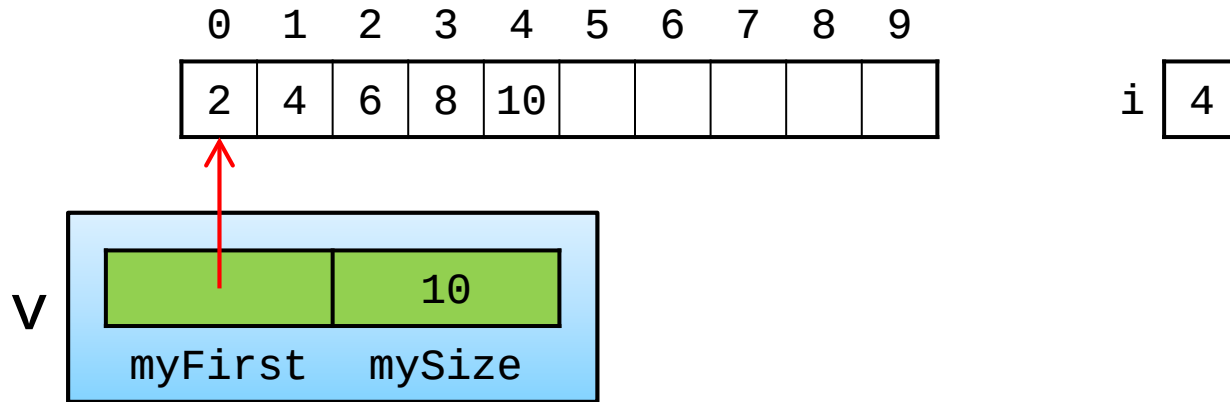
```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    cout < v.size() << endl;
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```



```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    cout < v.size() << endl;
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```



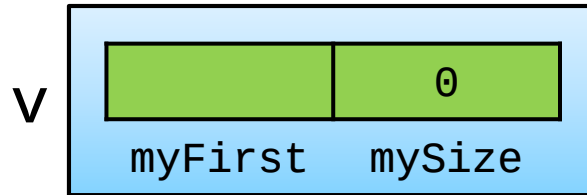
```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    cout < v.size() << endl;
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```



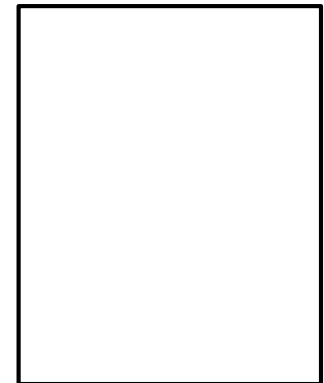
```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    cout < v.size() << endl;
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```

```
int main()
{
    const int vectorSize = 10;
    vector< int > v( vectorSize );
    for( int i = 0; i < vectorSize; i++ )
        v[ i ] = 2 + 2 * i;
}
```

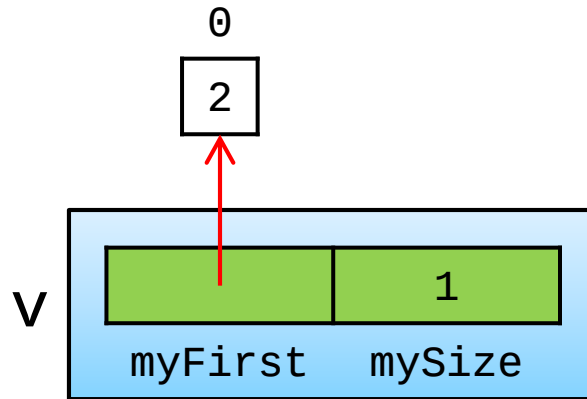
```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
        v.push_back( 2 + 2 * i );
}
```



```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
    {
        v.push_back( 2 + 2 * i );
        cout < v.size() << endl;
    }
}
```



Output

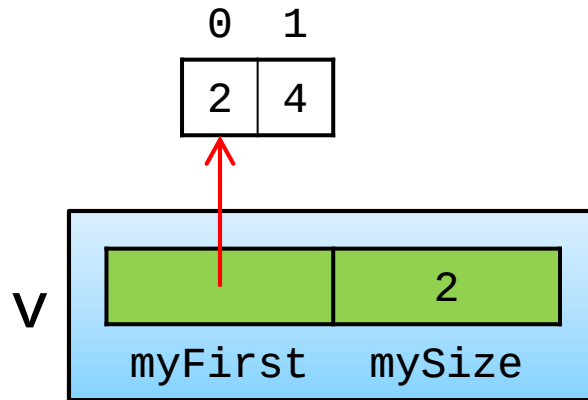


`i` 0

```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
    {
        v.push_back( 2 + 2 * i );
        cout < v.size() << endl;
    }
}
```

1

Output



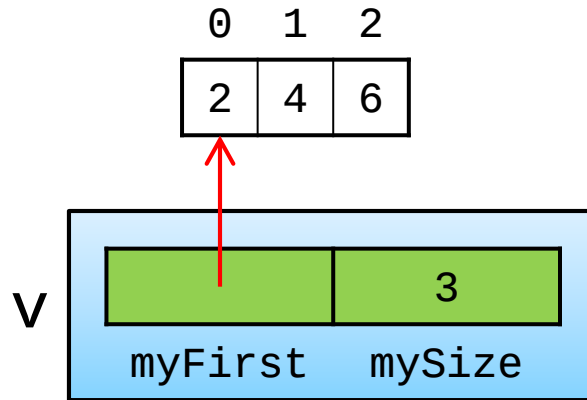
`i`

1

```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
    {
        v.push_back( 2 + 2 * i );
        cout < v.size() << endl;
    }
}
```

1
2

Output

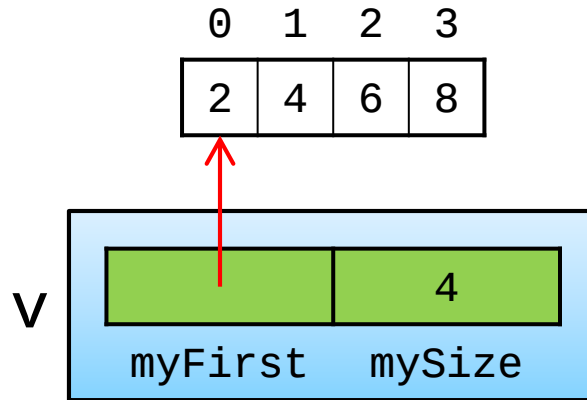


i 2

```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
    {
        v.push_back( 2 + 2 * i );
        cout < v.size() << endl;
    }
}
```

1
2
3

Output

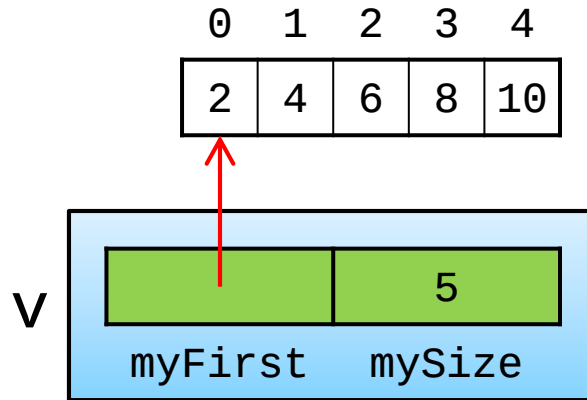


`i` 3

```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
    {
        v.push_back( 2 + 2 * i );
        cout < v.size() << endl;
    }
}
```

1
2
3
4

Output

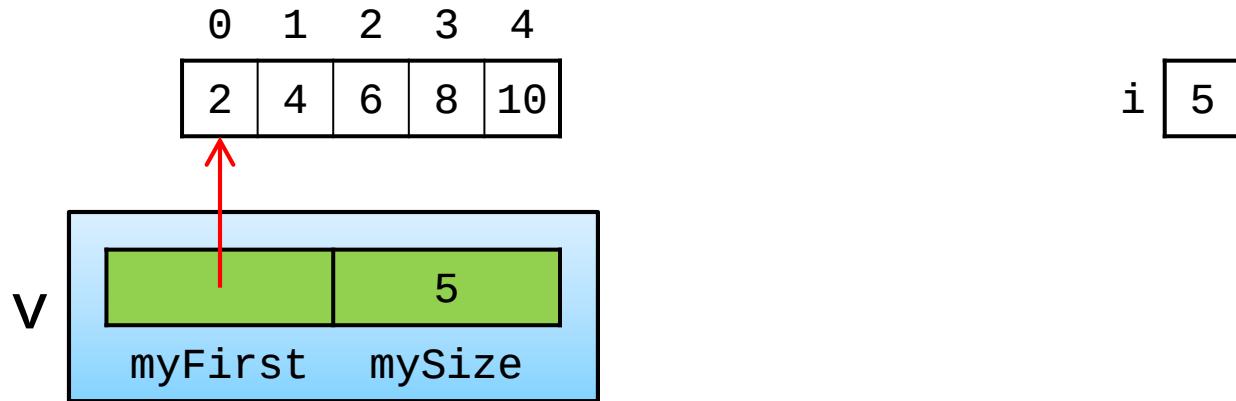


`i` 4

```
int main()
{
    const int vectorSize = 10;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
    {
        v.push_back( 2 + 2 * i );
        cout < v.size() << endl;
    }
}
```

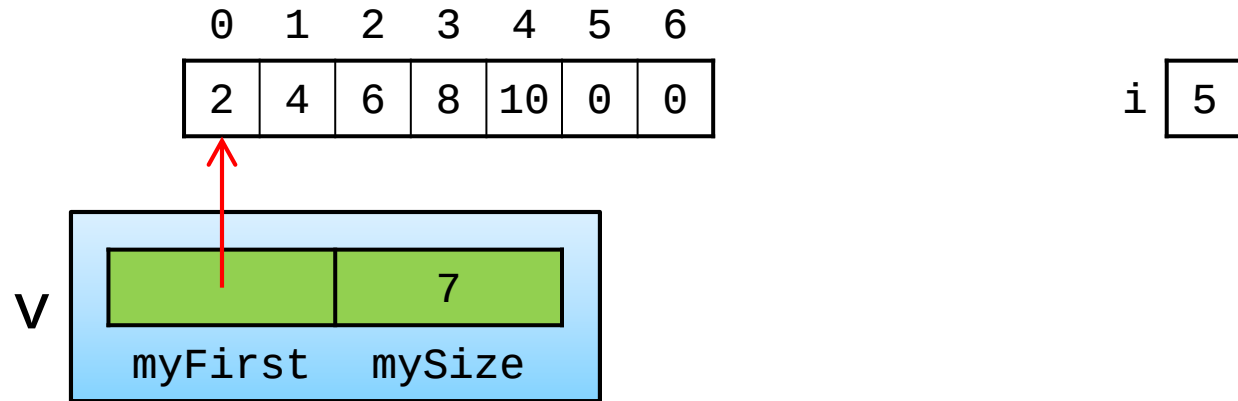
1
2
3
4
5

Output



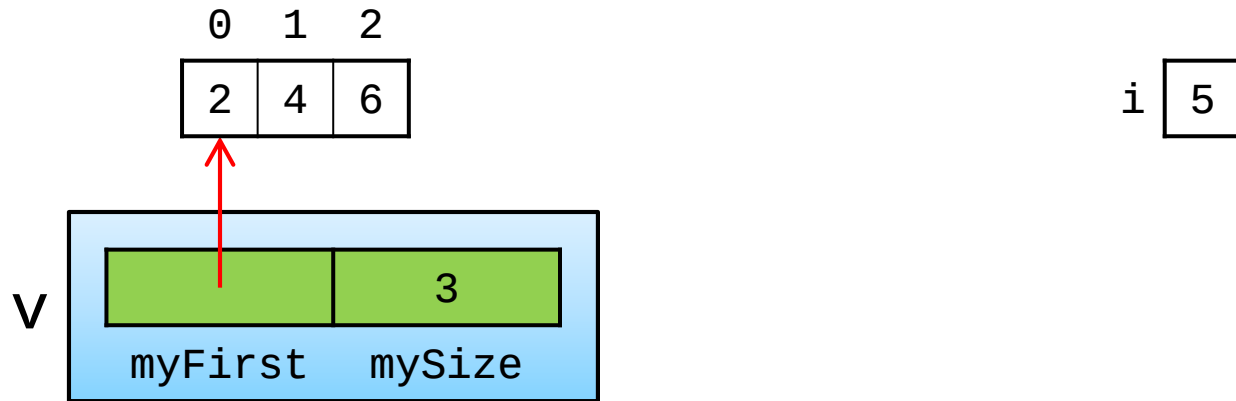
```
int main()
{
    const int vectorSize = 5;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
        v.push_back( 2 + 2 * i );

    v.resize( 7 );
    v.resize( 3 );
}
```



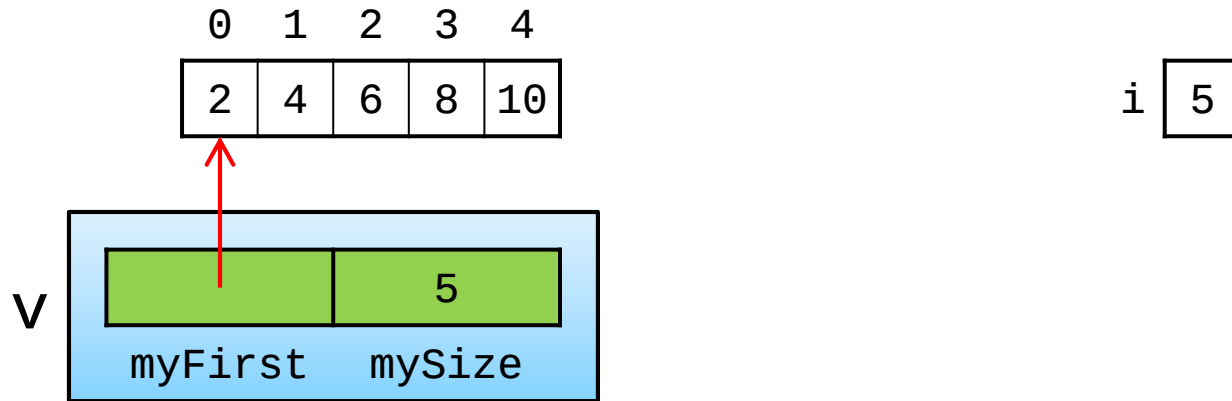
```
int main()
{
    const int vectorSize = 5;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
        v.push_back( 2 + 2 * i );

    v.resize( 7 );
    v.resize( 3 );
}
```



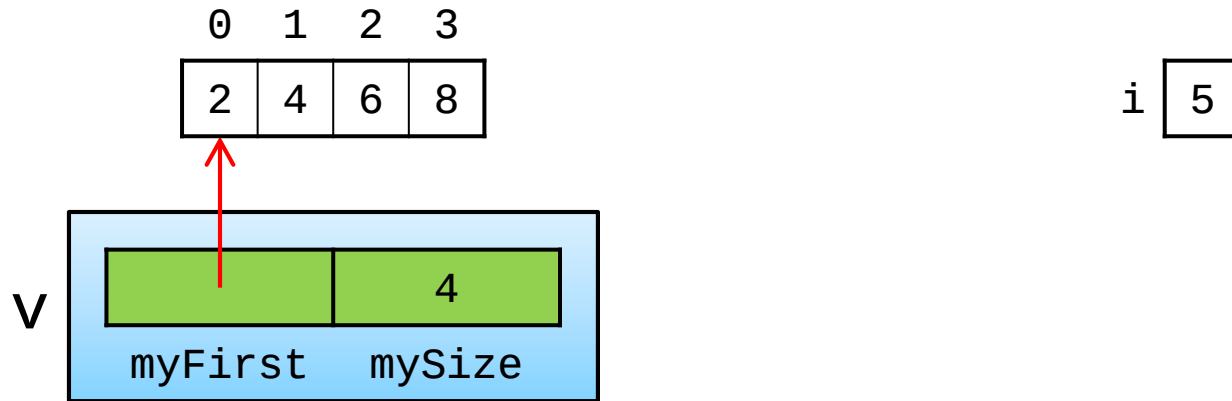
```
int main()
{
    const int vectorSize = 5;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
        v.push_back( 2 + 2 * i );

    v.resize( 7 );
    v.resize( 3 );
}
```

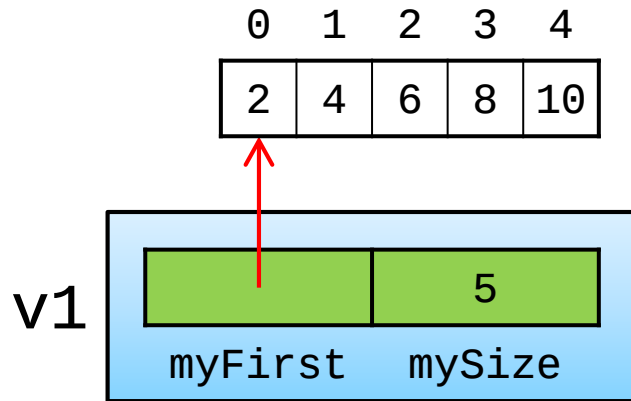
```
int main()
{
    const int vectorSize = 5;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
        v.push_back( 2 + 2 * i );

    v.pop_back();
}
```



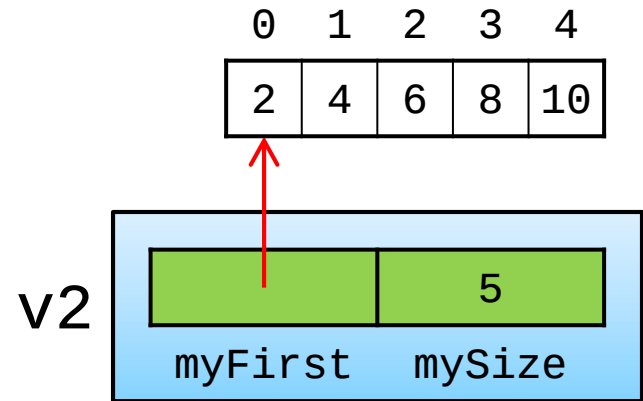
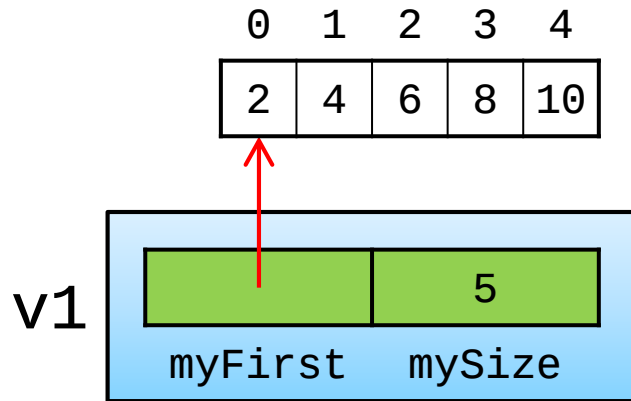
```
int main()
{
    const int vectorSize = 5;
    vector< int > v;
    for( int i = 0; i < vectorSize; i++ )
        v.push_back( 2 + 2 * i );

    v.pop_back();
}
```

```
int main()
{
    const int vectorSize = 5;
    vector< int > v1;
    for( int i = 0; i < vectorSize; i++ )
        v1.push_back( 2 + 2 * i );

    vector< int > v2( v1 );
}
```



```
int main()
{
    const int vectorSize = 5;
    vector< int > v1;
    for( int i = 0; i < vectorSize; i++ )
        v1.push_back( 2 + 2 * i );

    vector< int > v2( v1 );
}
```



```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ] = { 87, 68, 94, 100, 83, 78, 85, 91, 76, 87 };
    int total = 0;
    for( int i = 0; i < arraySize; i++ )
        total += a[ i ];
    cout << "Total of array elements: " << total << endl;
}
```

```
int main()
{
    const int arraySize = 10;
    int a[ arraySize ] = { 87, 68, 94, 100, 83, 78, 85, 91, 76, 87 };
    vector< int > v( a, a + 10 );
    int total = 0;
    for( unsigned int i = 0; i < v.size(); i++ )
        total += v[ i ];
    cout << "Total of array elements: " << total << endl;
}
```

```
int main()
{
    const int arraySize = 5;
    int a[ arraySize ] = { 0, 1, 2, 3, 4 };

    for( int i = 0; i < arraySize; i++ )
        cout << setw( 3 ) << a[ i ];

    modifyArray( a, arraySize );

    for( int j = 0; j < arraySize; j++ )
        cout << setw( 3 ) << a[ j ];
}

void modifyArray( int b[], int sizeOfArray )
{
    for( int k = 0; k < sizeOfArray; k++ )
        b[ k ] *= 2;
}
```



```
int main()
{
    const int arraySize = 5;
    int a[ arraySize ] = { 0, 1, 2, 3, 4 };
    vector< int > v( a, a + arraySize );

    for( unsigned int i = 0; i < v.size(); i++ )
        cout << setw( 3 ) << v[ i ];

    modifyVector( v );

    for( int j = 0; j < arraySize; j++ )
        cout << setw( 3 ) << v[ j ];
}

void modifyVector( vector< int > &v )
{
    for( unsigned int k = 0; k < v.size(); k++ )
        v[ k ] *= 2;
}
```

string

```
class string
{
    static const unsigned int npos = -1;

    unsigned int length()
    {
        return mySize;
    }

    void resize( unsigned int n )

    void clear()
    {
        mySize = 0;
    }

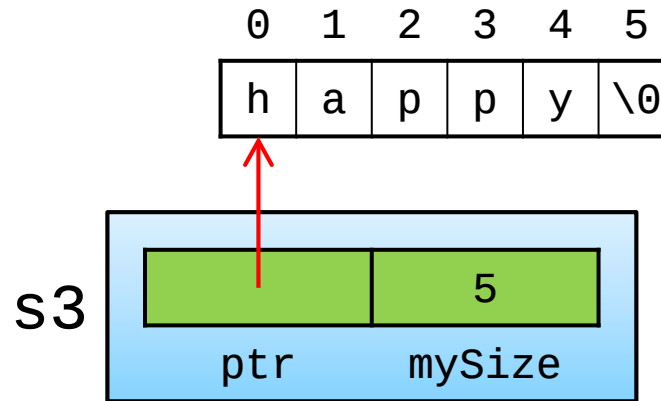
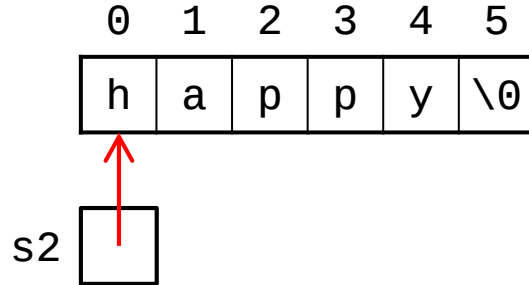
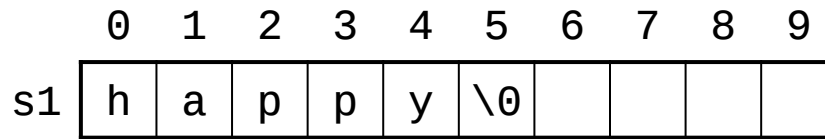
    bool empty()
    {
        return ( mySize == 0 );
    }
}
```

```
void push_back( char c )

void pop_back()
{
    mySize--;
}

string substr( unsigned int pos = 0, unsigned int len = npos )

unsigned int mySize = 0;
char *ptr = nullptr;
};
```



```
int main()
{
    char s1[ 10 ] = "happy";
    char *s2 = "happy";
    string s3( "happy" );
}
```

```

int main()
{
    string s1( "happy" );
    string s2( " birthday" );
    string s3;
    string s4;

    cout << "Enter a string: ";
    cin >> s4;

    cout << "s1 is \"" << s1 << "\"; s2 is \"" << s2
        << "\"; s3 is \"" << s3 << "\"; s4 is \"" << s4 << "\"\n\n";

    cout << "s1 length " << s1.size() << "; s2 length " << s2.size()
        << "; s3 length " << s3.size() << "; s4 length " << s4.size();

    cout << "\n\nThe results of comparing s2 and s1:"
        << "\ns2 == s1 yields " << ( s2 == s1 ? "true" : "false" )
        << "\ns2 != s1 yields " << ( s2 != s1 ? "true" : "false" )
        << "\ns2 > s1 yields " << ( s2 > s1 ? "true" : "false" )
        << "\ns2 < s1 yields " << ( s2 < s1 ? "true" : "false" )
        << "\ns2 >= s1 yields " << ( s2 >= s1 ? "true" : "false" )
        << "\ns2 <= s1 yields " << ( s2 <= s1 ? "true" : "false" );
}

```

```
cout << "\n\nTesting s3.empty():" << endl;

if ( s3.empty() )
{
    cout << "s3 is empty; assigning s1 to s3;" << endl;
    s3 = s1;
    cout << "s3 is \"" << s3 << "\"";
}

cout << "\n\nAfter s1 += s2, s1 is ";
s1 += s2;
cout << s1;

cout << "\n\ns1 += \" to you\" yields" << endl;
s1 += " to you";
cout << "s1 is " << s1 << "\n\n";

cout << "The substring of s1 starting at location 0 for\n"
    << "14 characters, s1.substr(0, 14), is:\n"
    << s1.substr( 0, 14 ) << "\n\n";

cout << "The substring of s1 starting at\n"
    << "location 15, s1.substr(15), is:\n"
    << s1.substr( 15 ) << endl;
```

```

string s5( s1 );
cout << "\ns5 is " << s5;

s1[ 0 ] = 'H';
s1[ 6 ] = 'B';
cout << "\n\ns1 after s1[0] = 'H' and s1[6] = 'B' is: " << s1;

cout << "\n\ns1[0] is " << s1[ 0 ] << "; s1[2] is " << s1 [ 2 ]
    << "; s1[s1.size()-1] is " << s1[ s1.size() - 1 ];

cout << "\n\nAttempt to assign 'd' to s1.at( 30 ) yields:" << endl;
s1.at( 30 ) = 'd';
}

```


Enter a string: Using class string

s1 is "happy"; s2 is " birthday"; s3 is ""; s4 is "Using"

s1 length 5; s2 length 9; s3 length 0; s4 length 5

The results of comparing s2 and s1:

s2 == s1 yields false

s2 != s1 yields true

s2 > s1 yields false

s2 < s1 yields true

s2 >= s1 yields false

s2 <= s1 yields true

Testing s3.empty():

s3 is empty; assigning s1 to s3;

s3 is "happy"

After s1 += s2, s1 is happy birthday

s1 += " to you" yields

s1 is happy birthday to you

The substring of s1 starting at location 0 for 14 characters, `s1.substr(0, 14)`, is:
happy birthday

The substring of s1 starting at location 15, `s1.substr(15)`, is:
to you

s5 is happy birthday to you

s1 after `s1[0] = 'H'` and `s1[6] = 'B'` is: Happy Birthday to you

`s1[0]` is H; `s1[2]` is p; `s1[s1.length()-1]` is u

Attempt to assign 'd' to `s1.at( 30 )` yields:

```
int main()
{
    string s1( "happy" );
    string s2( " birthday" );
    string s3;
    string s4;
    cout << "Enter a string: ";
    cin >> s4;
    cout << "s1 is \"" << s1 << "\"; s2 is \"" << s2
        << "\"; s3 is \"" << s3 << "\"; s4 is \"" << s4 << "\"\n\n";
}
```

```
int main()
{
    char s1[ 80 ] = "happy";
    char *s2 = " birthday";
    char s3[ 80 ] = "";
    char s4[ 80 ];
    cout << "Enter a string: ";
    cin >> s4;
    cout << "s1 is \"" << s1 << "\"; s2 is \"" << s2
        << "\"; s3 is \"" << s3 << "\"; s4 is \"" << s4 << "\"\n\n";
}
```

```
cout << "s1 length " << s1.length() << "; s2 length " << s2.length()  
    << "; s3 length " << s3.length() << "; s4 length " << s4.length();
```

```
cout << "\n\nThe results of comparing s2 and s1:"  
    << "\ns2 == s1 yields " << ( s2 == s1 ? "true" : "false" )  
    << "\ns2 != s1 yields " << ( s2 != s1 ? "true" : "false" )  
    << "\ns2 > s1 yields " << ( s2 > s1 ? "true" : "false" )  
    << "\ns2 < s1 yields " << ( s2 < s1 ? "true" : "false" )  
    << "\ns2 >= s1 yields " << ( s2 >= s1 ? "true" : "false" )  
    << "\ns2 <= s1 yields " << ( s2 <= s1 ? "true" : "false" );
```

```
cout << "s1 length " << strlen( s1 ) << "; s2 length " << strlen( s2 )  
    << "; s3 length " << strlen( s3 ) << "; s4 length " << strlen( s4 );
```

```
cout << "\n\nThe results of comparing s2 and s1:"  
    << "\ns2 == s1 yields " << ( strcmp( s2, s1 ) == 0 ? "true" : "false" )  
    << "\ns2 != s1 yields " << ( strcmp( s2, s1 ) != 0 ? "true" : "false" )  
    << "\ns2 > s1 yields " << ( strcmp( s2, s1 ) == 1 ? "true" : "false" )  
    << "\ns2 < s1 yields " << ( strcmp( s2, s1 ) == -1 ? "true" : "false" )  
    << "\ns2 >= s1 yields " << ( strcmp( s2, s1 ) >= 0 ? "true" : "false" )  
    << "\ns2 <= s1 yields " << ( strcmp( s2, s1 ) <= 0 ? "true" : "false" );
```

```
cout << "\n\nTesting s3.empty():" << endl;
if ( s3.empty() )
{
    cout << "s3 is empty; assigning s1 to s3;" << endl;
    s3 = s1;
    cout << "s3 is \"" << s3 << "\"";
}
cout << "\n\nAfter s1 += s2, s1 is ";
s1 += s2;
cout << s1;
```

```
cout << "\n\nTesting s3.empty():" << endl;
if ( strcmp( s3, "" ) == 0 )
{
    cout << "s3 is empty; assigning s1 to s3;" << endl;
    strcpy( s3, s1 );
    cout << "s3 is \"" << s3 << "\"";
}
cout << "\n\nAfter s1 += s2, s1 is ";
strcat( s1, s2 );
cout << s1;
```

```
cout << "\n\ns1 += \" to you\" yields" << endl;
s1 += " to you";
cout << "s1 is " << s1 << "\n\n";

cout << "The substring of s1 starting at location 0 for\n"
    << "14 characters, s1.substr(0, 14), is:\n"
    << s1.substr( 0, 14 ) << "\n\n";
```

```
cout << "\n\ns1 += \" to you\" yields" << endl;
strcat( s1, " to you" );
cout << "s1 is " << s1 << "\n\n";

char s6[ 16 ];
strncpy( s6, s1, 15 );
s6[ 15 ] = '\0';
cout << "The substring of s1 starting at location 0 for\n"
    << "14 characters, s1.substr(0, 14), is:\n"
    << s6 << "\n\n";
```

```
cout << "The substring of s1 starting at\n"  
    << "location 15, s1.substr(15), is:\n"  
    << s1.substr( 15 ) << endl;
```

```
char s7[ 10 ];  
memcpy( s7, s1 + 15, strlen( s1 ) - 15 );  
s7[ strlen( s1 ) - 15 ] = '\0';  
cout << "The substring of s1 starting at\n"  
    << "location 15, s1.substr(15), is:\n"  
    << s7 << endl;
```

```

string s5( s1 );
cout << "\ns5 is " << s5;
s1[ 0 ] = 'H';
s1[ 6 ] = 'B';
cout << "\n\ns1 after s1[0] = 'H' and s1[6] = 'B' is: " << s1;
cout << "\n\ns1[0] is " << s1[ 0 ] << "; s1[2] is " << s1 [ 2 ]
    << "; s1[s1.length()-1] is " << s1[ s1.length() - 1 ];
cout << "\n\nAttempt to assign 'd' to s1.at( 30 ) yields:\n";
}

```

```

char s5[ 30 ];
strcpy( s5, s1 );
cout << "\ns5 is " << s5;
s1[ 0 ] = 'H';
s1[ 6 ] = 'B';
cout << "\n\ns1 after s1[0] = 'H' and s1[6] = 'B' is: " << s1;
cout << "\n\ns1[0] is " << s1[ 0 ] << "; s1[2] is " << s1 [ 2 ]
    << "; s1[s1.length()-1] is " << s1[ strlen( s1 ) - 1 ];
cout << "\n\nAttempt to assign 'd' to s1.at( 30 ) yields:\n";
}

```